

Decisions, Debugging & Loops

Pre-Assessment 2 · 15 questions · 49 points

Name: _____ Date: _____ Score: / 49

This is a **pre-assessment**. It shows what you already know **before** we start this unit, so it is fine if some questions cover things you have not learned yet. Try every question you can. If you have no idea at all, write **“not yet”** and move on.

Work on your own, without a computer. You have about **45 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

1. True or False?

4 points

Write **True** or **False** for each comparison.

Comparison	Result	Comparison	Result
<code>7 > 3</code>		<code>10 != 10</code>	
<code>"cat" == "Cat"</code>		<code>4 <= 4</code>	

2. One equals or two?

2 points

(a) Which line correctly checks whether `score` equals 10? Circle one.

(1 pt)

☐ `score = 10`☐ `if score == 10:`☐ `if score = 10:`☐ `if score == 10`(b) In your own words, what does a single `=` do?

(1 pt)

3. Grade Calculator

4 points

```
if score >= 90:
    print("A")
elif score >= 80:
    print("B")
elif score >= 70:
    print("C")
else:
    print("Keep trying")
```

What does the code print when `score` has each value below?

score	Prints
85	
92	
70	
40	

4. Movie Night

4 points

```
if age >= 10 and has_ticket:
    print("Enter the movie")
if age < 8 or not has_ticket:
    print("Stay home")
if age > 10:
    if age < 13:
        print("Kid price")
    else:
        print("Teen price")
```

(a) What prints when `age = 12` and `has_ticket = True`?

(2 pts)

(b) What prints when `age = 15` and `has_ticket = False`?

(2 pts)

5. Name that error

4 points

Each program crashes. Write the error name: **SyntaxError**, **NameError**, **TypeError**, or **IndentationError**. Each one is used once.

Code	Error
<pre>print("hello)</pre>	
<pre>score = 5 print(scroe)</pre>	
<pre>print("Age: " + 12)</pre>	
<pre>if 7 > 3: print("big")</pre>	

6. Bug Zoo: Password Checker

3 points

This program has **three** bugs. Circle each bug and write the fixed program in the box.

```
password = input("Password: ")  
if password = "open sesame"  
    print("Welcome in!")  
else:  
    print("Access denied")
```


7. Rolling dice

2 points

```
import random
roll = random.randint(1, 6)
color = random.choice(["red", "blue", "green"])
```

(a) Which numbers can `roll` be? Circle one.

(1 pt)

☐ (A) 0 through 6

☐ (B) 1 through 5

☐ (C) 1 through 6

☐ (D) 1 through 7

(b) Name every value `color` could be.

(1 pt)

8. Countdown Rocket

3 points

Write exactly what this program prints.

```
n = 20
while n > 0:
    print(n)
    n = n - 6
print("Liftoff!")
```

9. Stop that loop!

2 points

```
count = 1
while count <= 5:
    print("Hello")
```

(a) What goes wrong when this runs?

(1 pt)

(b) Write the one line you would add (and show where) so it prints Hello 5 times.

(1 pt)

10. Which loop?

2 points

For each job, write **for** or **while**, and give a short reason.

(a) Print the 7 times table from 7×1 to 7×10 .

(1 pt)

(b) Keep asking the user for a word until they type `quit`.

(1 pt)

11. What does `range()` give?

4 points

List every number each `range` produces, in order.

Code	Numbers
<code>range(4)</code>	
<code>range(2, 7)</code>	
<code>range(1, 10, 2)</code>	
<code>range(10, 0, -3)</code>	

12. Counting and adding

4 points

(a) What does this print?

(2 pts)

```
word = "banana"
count = 0
for letter in word:
    if letter in "aeiou":
        count = count + 1
print(count)
```

(b) What does this print?

(2 pts)

```
total = 0
for i in range(1, 5):
    total = total + i
print(total)
```

13. Skip and stop

3 points

Write exactly what this program prints.

```
for n in range(1, 10):  
    if n == 3:  
        continue  
    if n == 6:  
        break  
    print(n)
```

14. Star Triangle

3 points

Write exactly what this program prints.

```
for row in range(1, 4):  
    line = ""  
    for col in range(row):  
        line = line + "*"   
    print(line)
```

15. Three Tries Password

5 points

Write a program that gives the user **up to 3 tries** to type the password `dragon`.

- If they type it correctly, print `Access granted` and stop asking.
- After a wrong try, print `Wrong, try again`.
- If all 3 tries are wrong, print `Locked out`.

END OF ASSESSMENT